

RWA tokenization: a step forward to invest in Ukraine

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Abstract. The article analyzes how tokenization can contribute to a more efficient use of state resources, reducing management costs and improving control over the use of budget funds. Studying the mechanisms for implementing tokenization is a key element of the study, as it will allow developing recommendations for creating a methodology for applying tokenization technology that will support innovation and ensure the protection of the interests of investors and local communities. The study aims to reveal the opportunities through which tokenization can contribute to the financial stability and growth of local budgets through the introduction of innovative financial mechanisms. The authors propose a principled logical architecture of a platform for tokenizing the assets of territorial communities and describe step-by-step the use case of issuing tokens from a state asset (RWA).

Keywords: tokenization, investments, local budget, territorial hromadas, assets, innovations.

Introduction

Ukraine has suffered significant losses as a result of full-scale aggression by the Russian Federation, covering various aspects of the functioning of the state and society. According to data [1], as of June 13, 2024, the total direct documented damage caused to residential and non-residential real estate, as well as other infrastructure of Ukraine as a result of military aggression, is estimated at \$95.5 billion, which is equivalent to approximately 2.6 trillion hryvnias at replacement cost. The structure of these losses is distributed as follows:

1. Residential buildings: 39% of the total (\$36.8 billion).
2. Infrastructure facilities: 33% (\$31.3 billion).
3. Business assets: at least \$8 billion (with a growing trend).
4. Agricultural sector: \$4.3 billion.
5. Public sector facilities: \$6.7 billion.

It is important to note that this estimate does not include the value of undamaged assets in territories occupied after 24.02.2022, as well as the value of assets occupied and/or damaged before this date. The geographical distribution of losses demonstrates the uneven impact of military actions on different regions of Ukraine. The most affected are the regions where hostilities were directly fought: Donetsk region - 25% of the total losses, Kharkiv region - 18%, Luhansk region - over 13%, Mykolaiv region - 9%, Zaporizhia region - 7%, Kyiv region - 7%, Chernihiv region - 6%. [1]

These data indicate the significant scale of destruction and economic losses inflicted on Ukraine because of military aggression. They emphasize the need to develop comprehensive recovery and reconstruction strategies that consider both national needs and the specifics of the most affected territorial hromadas. Recovery and reconstruction will require significant investments. At

the beginning of the full-scale aggression of the Russian Federation, local financial authorities faced many challenges, namely:

- instability of revenues during the budget year, which complicates expenditure planning;
- imperfect system of forecasting cash flows at the local level;
- delays in receiving transfers, which creates liquidity problems for local budgets;
- legislative restrictions on the possibilities of placing temporarily free budget funds, which reduce the flexibility of liquidity management;
- problems attracting investments due to the low investment attractiveness of the assets of territorial hromadas.

Encouraging investment plays a key role in the recovery and development of Ukraine's economy after significant losses caused by military aggression. Increased investment will lead to a multiplier effect on the economy, stimulating consumption, production and, as a result, economic growth. An active policy of attracting investment will signal the stabilization of the situation in the country and the restoration of international investors' confidence in the Ukrainian economy. It is the stimulation of investment that can ensure the inflow of capital necessary for the reconstruction of damaged infrastructure and the restoration of production capacities. Investments will contribute to the development of new industries and technologies, which will help diversify the economy of hromadas and reduce its dependence on traditional sectors that may have suffered as a result of the conflict, and will stimulate the creation of new jobs, which is critically important for the reintegration of internally displaced persons and the reduction of unemployment caused by military actions. Foreign investments are often accompanied by the transfer of technology and know-how, which contributes to the modernization of production processes and the increase of the competitiveness of the Ukrainian economy at the international level. Encouraging investments will contribute to the development of effective models of public-private partnership, especially in the field of restoration of critical infrastructure. Investments in scientific research and development (R&D) will stimulate innovative activity, which is a key factor of long-term economic development. For the effective implementation of the investment promotion policy at the level of territorial hromadas, it is necessary to ensure:

- stable and transparent legislative environment;
- an effective system for protecting investor rights;
- developed financial infrastructure, regulatory regulation of the implementation of the latest investment instruments and mechanisms, simplification of administrative procedures;
- targeted programs to support investors in priority sectors;

It is obvious that investment promotion can become a tool for the restoration and modernization of the Ukrainian economy, contributing not only to overcoming the consequences of

military aggression, but also to creating a basis for sustainable economic development in the long term. It is the development and implementation of the latest investment promotion tools that can become the driving force for changes in investment models at the level of territorial hromadas.

Tokenization of real assets is the creation of digital tokens that represent ownership or a stake in an underlying asset on a distributed network. This process has the potential to transform the global financial landscape by providing greater utility and the ability to compose assets. Tokenization of government assets involves converting physical or financial assets owned by a government into digital tokens on a blockchain, which can significantly improve the efficiency, transparency, and liquidity of these assets. This process creates a blockchain -based representation of the underlying instrument, reducing issuance and trading costs, reducing reliance on intermediaries, and providing greater transparency of the asset's lifecycle for all participants [1].

The issue of attracting investments and increasing local budget revenues is of constant interest to the scientific hromada. Certain aspects of this problem are highlighted in the works of M. Lutsiv [2], O. Dakhno [3], V. Yemts [4], Yu. Golynsky , V. Nurieva [5], A. Yakymchuk [6] and others. The authors of the article consider practical aspects of managing local budgets using software [7] and forecasting the flow of funds of local budgets using artificial intelligence [8]. To date, significant experience has been accumulated in scientific works on the use of asset tokenization. In particular, the studies of A. Andreev , D. Rudenko are devoted to this issue. [9], A. Martynova [10], I. Makalyuk [11], Ya. Belinska , S. Onyshko , R. Dyuka [12].

A review of the literature suggests that tokenization can protect sensitive data by replacing real values with virtual ones, making it a way to protect personal data from fraudsters [13]. A system for issuing, managing, and transferring asset-backed tokens typically includes a minting component and a commercial component that interacts with customers to control the issuance of tokens in their name [14]. For example, a method to facilitate the exchange of a stable cryptocurrency backed by government debt can be implemented through digital wallets and blockchain technology, ensuring secure and efficient transactions [15]. Tokenization can also solve the problem of infrastructure financing by increasing liquidity, transaction efficiency, and transparency between intermediaries, as demonstrated by the SEC-compliant energy security token Ziyen-Coin [16]. In the real estate sector, tokenization can increase liquidity and transparency by enabling fractional ownership and reducing transaction costs through smart contracts and blockchain technology [17, 18]. Securitization of assets, which involves packaging them to reduce risks or make them suitable for investment, can be simplified through tokenization, reducing the complexity, time, and cost of the process [19]. The tokenization method involves registering assets on a distributed ledger, assigning them with fungible or non-fungible tokens, and verifying ownership using public and private keys, which facilitates secure transactions [20]. However, the legal and regulatory framework for

tokenization is still developing, and countries such as Ukraine, Liechtenstein, and Switzerland are taking steps to provide legal clarity and protection for virtual assets and their transactions [20][21]. For example, Ukraine's draft law No. 3637 aims to define the legal status of virtual assets, provide legal protection for users, and allow cryptocurrency exchanges to officially operate, although it still raises questions about the classification and regulation of tokens [13]. Similarly, Liechtenstein and Switzerland have recognized the potential of distributed ledger technologies to perform the same functions as physical documents, with Switzerland focusing on DLT and securities law, and Liechtenstein seeking to create a holistic legal framework for the token economy [21]. Thus, while the tokenization of public assets offers numerous benefits, it requires careful consideration of legal and regulatory aspects to ensure security, transparency, and investor protection.

Existing scientific research does not address the issue of tokenization of state assets of territorial hromadas and the benefits it can provide in the context of attracting investment and increasing local budget revenues. The purpose of the article is to investigate the possibilities of tokenization of state assets to stimulate investment activity and increase revenues to local budgets. This study aims to assess the potential of tokenization for creating new financial instruments that will facilitate access to capital and stimulate economic development at the local level.

Our contribution

Digital assets streamline and automate bureaucratic processes, minimizing paperwork and reducing processing times. Combined with blockchain technology, this approach significantly increases transparency. The process of issuing and managing digital assets creates an immutable and public record of transactions, thereby reducing corruption and increasing accountability. Governments in many countries are using digital assets to provide public goods, ensuring that all hromadas have access to basic services, thus promoting financial inclusion.

These prospects are especially obvious for the public financial sector. Public finances require qualitative reform with the possibility of implementing mechanisms for fast and secure access to resources. Tokenization has become a breakthrough technology that changes our perception of assets and interaction with them. The development of tokenization platforms allows assets to be converted into digital tokens on the blockchain, which opens up opportunities for fractional ownership of assets, increasing their liquidity and accessibility. This process significantly lowers the barriers to entry for investors, allowing them to acquire and sell shares of assets without the need to manage a holistic object.

Consider the example of a high-value commercial property with an estimated value of \$10 million. In the traditional paradigm, such an investment would be out of reach for many individual investors due to the significant financial barrier to entry. However, the use of tokenization allows

for the distribution of ownership into 10,000 tokens, each worth \$1,000, significantly expanding the range of potential market participants. This democratization of access to investment opportunities leads to a significant increase in market liquidity. Investors can trade their tokenized shares on the secondary market, like securities transactions, which significantly simplifies the processes of entering and exiting investment positions compared to traditional transactions in the real estate market.

It is worth noting that increased liquidity is an attractive factor not only for individual investors, but also for institutional market participants. The latter can optimize the diversification of their portfolios by including smaller units of different asset classes, which together contributes to increasing the overall dynamics and efficiency of the market.

Blockchain technology is a fundamental foundation for providing increased transparency and security in the process of asset tokenization. A key characteristic of blockchain is the creation of an immutable transaction ledger, which makes it impossible to modify the recorded data later. This property plays a critical role in building trust in investment processes, especially in sectors characterized by a high level of complexity and opacity.

Consider the real estate industry as an example. Transactions in this sector typically involve multiple parties, extensive documentation, and regulatory scrutiny. Blockchain significantly simplifies this complexity by providing a single, immutable record of every transaction from the initial offering to every subsequent trade. Each token represents a stake in an asset, and every transaction with those tokens is recorded on the blockchain. This transparency not only makes it easier to track ownership but also minimizes the likelihood of disputes over ownership of assets.

The security aspect of blockchain is based on its decentralized architecture, which provides protection against fraud and unauthorized access. Unlike traditional centralized databases, which create a single point of failure and vulnerability to attacks, blockchain distributes data across a network of computing nodes, which makes it much more difficult for attackers to compromise the integrity of the data. Additionally, the use of cryptographic methods in blockchain ensures reliable encryption of all transactions, increasing the level of security of tokenized assets.

This enhanced level of transparency and security is particularly attractive to markets where issues of fraud, misrepresentation, and mistrust are prevalent. By ensuring the transparency and immutability of every transaction, blockchain technology not only minimizes these risks, but also creates a foundation for establishing trust and accountability that has previously been difficult to achieve in certain asset classes.

Thus, the integration of blockchain technology into the asset tokenization process represents a significant step forward in increasing the efficiency, security, and transparency of investment transactions, especially in sectors with a high level of complexity and traditional opacity.

tokenization demonstrates significant potential to improve transaction profitability by streamlining processes and minimizing or eliminating the need for traditional intermediaries such as brokers, agents, and legal advisors. This transformation has a particularly pronounced impact in high-value asset transactions, where brokerage fees traditionally account for a significant portion of costs.

Smart contracts are self-executing software protocols in which the terms of a contract are directly incorporated into the software code. These contracts ensure automatic compliance and execution of the terms of the contracts, significantly reducing the need for legal intermediaries and significantly speeding up the transaction process.

Moreover, transparency, which is an inherent characteristic of blockchain technology, minimizes the need for comprehensive legal due diligence (diligence), which traditionally increases both financial costs and time costs. Investors get the opportunity to track the full history of the token directly on the blockchain, including data on previous owners and completed transactions, which provides an additional level of trust without charging additional costs.

Our approach is based on the analysis of international experience in the implementation of digital assets. Estonia, a world leader in digital governance, uses blockchain technology to create and protect digital identities, electronic public registries, demonstrating innovative use of digital assets at the national level. One approach to tokenizing city assets was demonstrated by the city administrations of Miami and New York. They tokenized city assets and created City projects Tokens, which allows people to invest in tokens tied to cities. MiamiCoin generates revenue, part of which is transferred to the city for the development of infrastructure and social projects. This has allowed for significant funds to be used to develop city initiatives, demonstrating the benefits of transparency, decentralization, and public involvement in the development of their hromadas.

One of Argentina's largest banks, BBVA, has successfully implemented a commercial real estate tokenization project, which has enabled investors to purchase shares of these assets in the form of tokens. This has lowered the threshold for entry into highly liquid assets and attracted new investors, increased liquidity in the real estate market as a whole, significantly reduced transaction costs, and ensured transparency in asset ownership and management.

Smartlands project (UK) tokenized student housing, allowing investors to purchase shares in the property through tokens. The tokenization platform uses the blockchain to issue tokens that represent ownership shares in student housing projects. Investors receive rental dividends and the ability to resell the tokens on the secondary market, ensuring transparency, lowering the threshold for investment, and increasing the liquidity of typically illiquid assets.

Propeller and Fluidity platforms have implemented a real estate asset tokenization service in New York using smart contracts to tokenize property, allowing investors to obtain shares of real

estate and significantly save on commissions, which are considered among the highest in the real estate market.

International experience shows that tokenization of hromada assets can be an effective tool for financing and managing assets, while ensuring transparency, accessibility and liquidity. Successful examples of projects in many countries around the world demonstrate the potential of this technology to attract investment and develop hromada projects.

Given the advantages analyzed above, we propose a basic logical architecture of a platform for tokenizing the assets of territorial hromadas (Fig. 1).

In Fig. 1 the following abbreviations are used:

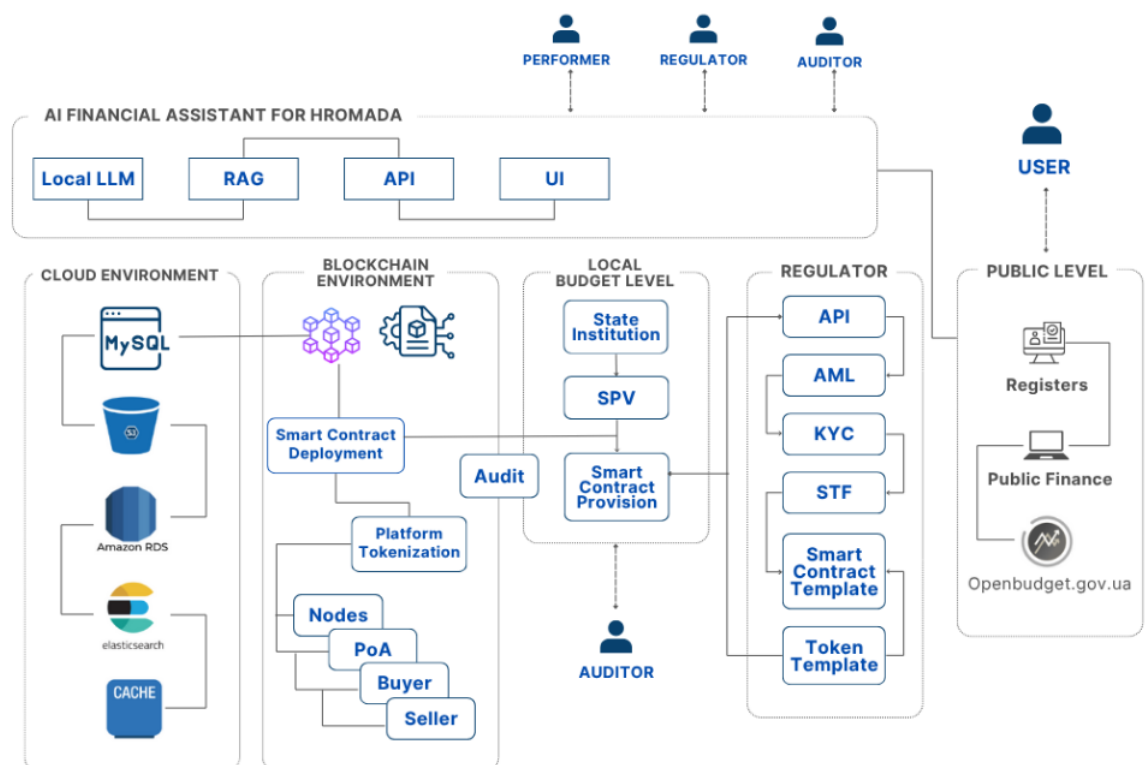


Fig. 1. Basic logical architecture of the platform for tokenization of territorial hromada assets*

*Source: created and proposed by the authors

API (Application Programming Interface) Program Interface or application programming interface) - which allows components to integrate and interact with each other

AML (Anti -Money Laundering) is a set of measures, procedures and regulatory requirements aimed at preventing and detecting money laundering and terrorist financing;

KYC (Know Your Customer) Your Customer or “know your customer”) is the process of verifying the identity of customers by financial institutions and other entities to prevent money laundering, terrorist financing, fraud and other illegal activities; CFT (Combating the Financing of Terrorism or combating the financing of terrorism) is a set of measures and regulatory requirements aimed at preventing and detecting the financing of terrorist activities;

SPV (Special Purpose Vehicle, or in the context of tokenization, special purpose vehicle) is a separate legal entity created to achieve a specific financial goal, such as isolating financial risks or facilitating the financing of projects by a territorial hromada;

PoA (Power of Attorney, or in the context of tokenization, a digital power of attorney mechanism) - a mechanism of powers of attorney for the rights to represent the interests of investors in a digital system;

Amazon RDS is a distributed relational database;

Local LLM (Local Large Language Model or local large language model) is a set of artificial intelligence algorithms used to understand and generate natural language;

RAG (Retrieval-Augmented Generation or search-augmented generation) is a method in natural language processing that combines technologies for selecting relevant information and text generation.

We propose a general approach of issuing tokens from a government asset (RWA) as follows.

1. The first step is to identify and value the asset. Experience suggests that this will be implemented in the form of registries, from which information will be available for further actions, such as conducting an independent expert assessment of the asset's value and analyzing the legal status and potential restrictions on tokenization.

2. In the second step, it is proposed to implement all the necessary economic and legal compliance, namely, the creation of a legal basis for tokenization in accordance with current legislation (creation of a digital identity of a legal entity that owns the rights to the object of tokenization; definition of the rights and obligations of token owners; requirements for AML, KYC and CFT; development of mechanisms for interaction between the state and investors and other regulated rights and mandatory norms that will become part of the smart contract.

3. The third step will be to analyze and select a suitable blockchain platform (e.g. Ethereum, Polkadot, Cardano), assess scalability, security, and compatibility with existing financial systems. The state will likely be interested in creating its own blockchain platform, given that digital transformation is recognized as one of the top priority reforms - there are many potential services and economic relations that can be digitized and regulated using blockchain mechanisms.

4. In the fourth step, a smart contract ready for provisioning will appear from the template. To this end, the main provisions of the smart contract will be developed, defining the functionality of the tokens; revenue distribution, governance, and liquidity mechanisms will be implemented; and an audit will be conducted.

5. In the fifth step, the total number of tokens, their nominal value and the procedure for the initial token issuance (Initial) will be determined. Token Offering, ITO).

6. The sixth step should involve the registration and settlement of tokens issued for regulatory purposes: obtaining the necessary permits from regulatory authorities for trading, registering tokens as securities (if this is proposed through this mechanism).

7. Next will be a series of administrative, organizational and technical tasks that embody the life cycle of the overall system, namely: development or integration with existing platforms for secondary token trading; ensuring liquidity and transparency of pricing; implementing a system for token owners to participate in making decisions on asset management (it would probably make great sense to implement such management due to the priority of hromada members for investing); developing protocols for distributing future income and expenses; implementing multi-level security systems to protect tokens and related transactions; regulatory and legal regulation of conducting independent audits to ensure transparency and trust; creating a system of regular reporting for token owners and constant monitoring of the effectiveness of asset use and compliance with regulatory requirements.

The implementation of the proposed approach requires close cooperation between government agencies, legal experts, technology specialists and financial consultants. It is important to note that the specific steps may vary depending on the specifics of the asset, the legislative framework and the chosen technological platform. The analysis demonstrates that this approach should be considered as the basis for a future strategy for using tokenization technology as a tool for providing liquidity in local hromadas. The strategy should include regular review and updating of smart contracts in accordance with changes in legislation, market needs, scaling the project horizontally (within hromada assets) and vertically (within state assets).

The analysis of challenges provides a complete picture of the complexity of implementing tokenization technology. However, if these difficulties are successfully resolved, tokenization technologies will significantly contribute to increasing interest in investing in local hromada assets. Such challenges include:

1. The technological infrastructure required for asset tokenization is complex. The proposed principle logic diagram for implementing the tokenization concept provides an idea of the general approach without revealing the complex details of the implementation. The solution to such difficulties is due to the need to develop a public-private partnership with the involvement of commercial enterprises that have gained sufficient experience in implementing tokenization projects.

2. The regulatory framework for the tokenization of government assets is not well developed and requires further research. Currently, proactive legislative reviews and regulatory adjustments are needed to fully legalize and protect tokenized assets. Regulating tokenized assets can be complex

due to their decentralized nature. It is important to create a clear regulatory framework that ensures transparency, investor protection, and financial stability.

Conclusion

Thus, the implementation of asset tokenization based on blockchain technology represents an innovative approach to optimizing investment processes, characterized by the potential to significantly increase the efficiency of operations and reduce transaction costs in various sectors of the economy. From the analysis, the use of tokenization technology at the level of asset management of territorial hromadas has great transformative potential, creating the prerequisites for a revolution in approaches to investing, providing public services and attracting capital for the development and restoration of hromadas. Asset tokenization will provide local hromadas with the potential to:

- increase in revenues to local budgets;
- optimization of local taxes and fees by expanding the tax base and reviewing the rates of local taxes and fees, and their differentiation according to economic activity in the region;
- introduction of new types of local taxes that meet the modern needs of the hromada;
- improving tax administration by involving the population in tax payment control through the creation of transparent reporting and control mechanisms;
- creating favorable conditions for attracting investments by lowering the entry threshold for investing in highly liquid assets and providing privileges and priorities to local hromadas, through infrastructure development, tax breaks, and support for innovation.

tokenization infrastructure, local financial institutions will receive increased asset liquidity, increased transparency and security, simplified investment processes, efficient asset management, attraction of additional investments, reduced transaction costs, and additional mechanisms to stimulate the development of the local economy.

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